

China and Universals
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Introduction

Western scholars have looked to China to provide answers to age-old questions about the nature of universals. For our purposes, a “universal” is some idea that may occur to any member of any culture at any time and applies to everyone, everywhere. We cannot determine if an idea is truly universal if we have not tested the idea in another culture. If Chinese culture, which developed without influence from the Judeo-Christian West, accepts an idea similar to a Western idea, we may accept the idea as a universal notion, or so the theory goes. Universals are ideas that “ought” to occur to all people, everywhere, at some time in their cultural history without borrowing the idea from other cultures. Scholars talk about truth, beauty, compassion, logic, even grammar, as universals. Societies that do not contain these ideas or do not contain them in the same way as does the West are viewed as somehow defective. Universals are, after all, universal. Any insights that these defective societies may have are often discounted.

I question the idea of universals, whether they exist and, if they do, how we can identify them. Obviously, there is not enough room in this paper to completely answer the question so I limit my comments to a partial answer of the second question: how can we identify universals, assuming that they exist? I raise the question using Leibniz’s claim that the hexagrams of the *I Ching* are a form of binary mathematics.

Leibniz and the *I Ching*

The German philosopher Gottfried Leibniz corresponded with a number of Jesuit missionaries to China, one of whom, Fr. Bouvet, sent him a copy of a hexagram table from the *I Ching*¹. Bouvet became familiar with Leibniz’s development of binary mathematics and noticed a similarity between it and the hexagrams. For those not familiar with binary mathematics, the core of the system is the representation of all numbers

using only 0 and 1². A binary number system has the ability to encode any number, albeit in a cumbersome way. Binary numbers were a curiosity for several hundred years until the advent of the electronic computer made the use of the system necessary³. Leibniz saw binary numbers as a basic way, perhaps an ancient way, of communicating knowledge. From his earliest days, Leibniz wanted to construct what he called a “Universal Characteristic”, a sort of language *in which all notions and things are nicely ordered, a language with whose help different nations can communicate their thoughts, and each, in its own language, read what the other wrote*, Ariew and Garber, 1989:6. Leibniz never completed his work on a Universal Characteristic but he made some comments about it throughout his writings. Leibniz did not propose the creation of a language like Esperanto but, rather, a sort of mathematical language. Leibniz wanted to calculate knowledge in the same way that one calculates mathematics or proves logical theorems⁴.

The Universal Characteristic was not a random musing but a logical outcome of Leibniz’s worldview. Leibniz believed that there was a necessary reason for every single thing in this best of all possible worlds. There was little or no room for chance in his worldview. Everything was inherently knowable and knowable in the same way. Algebra as we know it today had been newly invented when Leibniz was young and was, to use a modern computer analogy, the “killer app”⁵ of its day. He invented the calculus and at least stated the basis for symbolic logic. To Leibniz, notation was important. Leibniz saw a simple workable notation as the key to his Universal Characteristic. As Davis tells us:

The symbols j and d did not represent meaningless sounds like the letters of a phonetic alphabet; they stood for concepts and thus provided a model for Leibniz’s boyhood wonderful idea of an alphabet representing all fundamental concepts. Davis, 2000:12

Leibniz’s notion of a Universal Characteristic was not the pipedream of an isolated philosopher. Sinology and linguistics were just beginning during Leibniz’s lifetime. There was a belief that a *Clavis Sinica* (a “key to Chinese”)

could be found or developed that would speed up the acquisition of Chinese by European scholars. Some believed that there was a “Primitive Language”, given by God to Adam and that Chinese may have been that language. The Primitive Language was universal. Others believed that the Primitive Language could not be recovered but that a new equivalent could be created using “Real Characters”. *Real Characters involved writing that represents not merely letters or words but things and ideas. Since Real Characters are not arbitrary signs, they need not be learned but rather convey their meaning in a manner that is universally understood* Mungello, 1999:63.

Because of his world-view, Leibniz thought that knowledge about the world could be discovered, forgotten, and rediscovered. He thought that his binary number system was a rediscovery of a system known to the ancient Chinese. In his *Discourse on the Natural Theology of the Chinese* (hereinafter *Discourse*), he says:

It is indeed apparent that if we Europeans were well enough informed concerning Chinese Literature, then, with the aid of logic, critical thinking, mathematics and our manner of expressing thought – more exacting than theirs – we could uncover in the Chinese writings of the remotest antiquity many things unknown to modern Chinese and even to other commentators thought to be classical. Reverend Father Bouvet and I have discovered the meaning, apparently truest to the text, of the characters of Fohi⁶, founder of the Empire, which consist simply of combinations of unbroken and broken lines, and which pass for the most ancient writing of China in its simplest form. Cook and Rosemont, 1994:132-133.

Leibniz talks about the *I Ching* hexagrams as though they were a form of writing but later he makes it clear that he thinks that the hexagrams are a form of mathematics. I assume that by “writing”, Leibniz really means “notation”.

Leibniz makes several claims in this work. First, he makes the unsupported claim that the contemporary Chinese did not know or

understand their *writings of the remotest antiquity*. He then claims that, if we only had the texts, we could learn the correct interpretation of those texts by using *logic, critical thinking, mathematics and our manner of expressing thought*. Thirdly, he claims that the hexagrams are *the most ancient writing of China in its simplest form*. Of course, he also claims to have *discovered the meaning, apparently truest to the text, of the characters of Fohi, founder of the Empire*.

Recall that Leibniz lived during the time of the Rites Controversy, part of which had to do with the proper translation for “God”. Leibniz, although not Catholic, sided with the Jesuits in this controversy. Leibniz thought that although the Chinese did not receive revelation in the same way that the West did, the ancient Chinese were able to deduce the basic truths about “God” because those truths are universal, therefore, comprehensible by rigorous thought. Leibniz was, for whatever reason, convinced that the pre-Confucian sages had in fact done so. In his *Remarks on Chinese Rites and Religion*, (hereinafter *Remarks*) he says:

And thus, as far as I understand, I think the substance of the ancient theology of the Chinese is intact and, purged of additional errors, can be harnessed to the great truths of the Christian religion. Fohi, the most ancient prince and philosopher of the Chinese, had understood the origin of things from unity and nothing, i.e., his mysterious figures reveal something of an analogy to Creation, containing the binary arithmetic (and yet hinting at greater things) that I rediscovered after so many thousands of years, where all numbers are written by only two notations, 0 and 1. Cook and Rosemont, 1994:73

Since this is so, he claimed, the Chinese can now see that the Western God and Shang Di are one and the same; the Chinese will accept the Christian faith and peace, harmony, and brotherhood will break out between China and the West. Whether or not we see Leibniz and many of his contemporaries as deluded, it is important to understand that the basis of their strong beliefs in

this project was religious: that a divine being, in one moment, created the world according to a plan. A plan suggests order; order suggests reason; reason suggests that the plan can be understood (at least in part) by human reason. Leibniz believed that the human ability to understand the divine plan was part of the divine plan.

Leibniz did not think that reasoning was culture-specific. All human beings everywhere could reason⁷. Because of his fascination with notation and his strong belief that suitable notation makes suitable systems, Leibniz seems to have confused the ability to reason with the method of reason. Leibniz developed the notation for calculus that we know today. He had every reason to believe that an elegant notation indicated a well-thought out system of reasoning.

What Leibniz saw was a hexagram system that used two symbols like his binary mathematics. Instead of 0 and 1, the hexagrams used a broken (yin) and a solid (yang) line. Logically, the numbers and the lines are equivalent. In the chart sent to Leibniz, the hexagrams are arranged in order from 0 to 63 only if binary numbers are derived from them in Leibniz's fashion. We can still see Leibniz's handwriting that notes the Arabic numeral corresponding to each hexagram written above it. The chart is a double set of hexagrams. One set, arranged in a 8 x 8 square, is centered within a circular hexagram set. The sequence of hexagrams, also hand numbered, in the circle is not in numerical order. The chart is printed on two pages. The right-hand page⁸ has a hand numbered "27" in its lower right-hand corner. The left-hand page has an upside-down "28" in the upper left-hand corner. Using Leibniz's handwritten numerals, page 27 contains hexagrams 0 to 31, with 0 at the bottom and 31 at the top. Page 28 starts with 32 on the bottom, next to 0, and goes to 63 on the top, next to 31.

Leibniz read the hexagrams in the square from the top down. He read the hexagrams in the circle from the outside of the circle towards the center. The equivalence between the order of hexagrams in the chart and the order of integers only holds because Leibniz "misinterpreted" it⁹. Leibniz did not comment on the difference between the order of the square and the order of

the circle. There is a startling similarity between the use of two symbols in the hexagrams and Leibniz's binary numbers that is highlighted by the order of the square. It is easy to see how working out the numerical relations between the hexagrams and Arabic numbers excited Leibniz. He was so excited by the similarity that he overlooked many dissimilarities.

Numerical versus symbolic systems

Leibniz, possibly influenced by Bouvet, Mungello, 1999:69, dismisses the notion that the hexagrams are symbolic and not mathematical. He said, in the *Discourse*:

So it seems that Fohi had insight into the science of combinations, but the Arithmetic having been completely lost, later Chinese have not taken care to think of them in this [arithmetical] way and they have made of these characters of Fohi some kind of symbols and Hieroglyphs, as one customarily does when one has strayed from the true meaning... Cook and Rosemont, 1994:133-134

Nothing in the *Discourse* or in the other works translated in Cook and Rosemont clarifies what Leibniz meant by *the science of combinations*. Probably he means "binary arithmetic" since 0s and 1s or broken lines and solid lines have to be combined to make numbers and hexagrams.

When we think of logic or mathematics, we think of highly abstract systems used to understand the world. These systems use symbols. Leibniz's dichotomy symbols : mathematics raises the question of the difference between the two. While his works on China translated in Cook and Rosemont do not contain Leibniz's explanation of this dichotomy, we speculate that the difference has to do with the precision of reference that each may have. Mathematics, on the other hand, always strives for precision. Recall Leibniz's goal for his Universal Characteristic. He wished to construct a language that represented universal notions that were understandable by everyone who understood the language. There was no room for ambiguity in this project. Each "word" of the language could have only one meaning. Symbols could hold more than one meaning or could be so ill defined that there could be

arguments or confusion about what they represented. Symbols would not do for his Universal Characteristic. Looked at from this point of view, Leibniz thought that he found a notation (hexagrams) of a form (binary mathematics) of universal knowledge (known to Fu Xi) that was forgotten and misinterpreted (rendered symbolically) by later generations of Chinese scholars. Leibniz rediscovered the “real” meaning of the hexagrams and wished to restore them to their original purpose.

Leibniz was well aware that Europe had not acquired a substantial portion of the corpus of Chinese texts. Reading the first quoted passage closely, Leibniz seems to have believed that more texts containing hexagrams like the *I Ching* would emerge from China. The *Discourse* was never completed. Leibniz died shortly after he had written what we have. We do not know whether Leibniz would have continued in his belief that the hexagrams were a form of binary arithmetic after failing to learn of other texts like the *I Ching*.

Place value systems and hexagram rotations

Binary numbers have a strict place value order. They begin at the right and increase in value as one moves left, each place holding a value double that of its neighbor to the right. For example, the binary number 1101 can be written in Arabic numerals as 13. Reading from the right to the left of the binary number, there is a 1 in the units place (there is one 1 in the number), a 0 in the twos place (no 2s), a 1 in the fours place (one 4), and a 1 in the eights place (one 8). We add the places with 1s ($8 + 4 + 1$) to get 13. A hexagram is a stack of lines with no place value order. Leibniz “rotated” the hexagrams 90° clockwise to get a place value order:



Figure 1. Leibniz’s Hexagram Rotation

The hexagram in **Figure 1**, Weiiji¹⁰, becomes the binary number 010101¹¹ or 21. To get the traditional Chinese reading, from the bottom of the hexagram and reading up, rotate the hexagram 90° counterclockwise:



Figure 2. Traditional Chinese Reading

Thus rotated, Weiji's binary number is 101010 or 42. Not all rotations produce one number that is double of the number produced by the other rotation¹². Sometimes rotations do not affect the numbers at all, e.g., Jian. Sometimes the numbers have no obvious¹³ relation, e.g., Tongren's Leibniz rotation produces 47 and its Chinese rotation produces 61.

A numeral in a particular place means one and only one thing¹⁴. However, there is no unique place value order for hexagrams. Hexagrams can be rotated either way to produce a binary number; neither rotation is incorrect. A valid binary number is produced by both rotations. There is no way for us to tell whether either number is the "correct" number. The binary number produced by a rotation may be "read back" into a "valid" hexagram but the hexagram produced may not be the hexagram that was initially rotated. For example, if the hexagram Guai is rotated clockwise to give the binary number 111110 (top yin line and 5 yang lines) and is then read back in the Chinese fashion, we get the hexagram Gou (five yang lines with a bottom yin line)¹⁵. This ambiguity is strong evidence that the hexagrams are not intended to be mathematical or, at least, mathematical in the Western sense.

The hexagram set

In the *Discourse* and, by implication¹⁶, in the *Remarks*, Leibniz expected the hexagram sequence to be expandable, at least in principle, to *all the natural numbers in order*, Cook and Rosemont, 1994:137. The hexagram set is closed, i.e., the hexagrams set cannot produce all natural numbers. To do so, the yin yang line notation needs to be expanded to include more lines. Such an expansion would be highly unwieldy and unlikely to be used. Leibniz's binary arithmetic can produce all natural numbers and is, therefore, open.

Human beings have little use for binary arithmetic in their daily affairs. For humans, binary arithmetic is not as simple as our usual denary form but there are some advantages. On the plus side, binary arithmetic requires the

least number of characters that a mathematical system can use¹⁷. The trade-off is the sheer bulk of the representations. Numbers written in binary are cumbersome. The lack of variation in characters taken together with the length of a binary number creates substantial opportunity for human error. A number in binary notation takes up more physical room than a number in Arabic numerals; compare 1000000 with 64. In binary, every order of 2 adds one more digit to the left end of a binary number, e.g. 128 is 10000000. What about hexagrams? A hexagram treated as a binary number cannot reproduce numbers larger than 63. The top-most line in a hexagram is the 32s place; adding another line above it would represent the 64s place. We now have a septagram. For convenience, let us call a notation system using yin and yang lines to express numbers a *yyGram system* and, when speaking of any general member of that system, a *yyGram*. We may replace the *yy* with a prefix indicated the number of lines in an individual or group of *yyGrams*. The number of lines (*yy*) in a *yyGram* will be referred to as the *size* of the *yyGram*.

Any left-most binary digits that are zeros may be omitted, e.g., 4 may be written 100 rather than 0100 or 00000100. What about septagrams? Will all septagrams have to display the seventh line, even if it is a yin, i.e., broken, line? An examination of the hexagram chart shows us that the hexagrams do precisely this. All “digits” must be expressed, whether or not they are leading yin lines. In other words, all *yyGrams* have the same number of lines as any other *yyGrams* of the same size, regardless of the number of leading yin lines¹⁸. This simple little fact creates a problem. The binary number that represents 10,000 is 10011100010000. It contains 14 characters and would be expressed by 14 lines in our hypothetical expansion of the *yyGram* system. The same number can be expressed by one traditional character, *wàn*. There is no benefit and considerable disincentive to the use of *yyGrams* for something as simple as counting.

The *yyGram* system that we see in Chinese literature is limited to trigrams and hexagrams¹⁹. A *yyGram* system limited to trigrams and hexagrams is bounded by 0 and 63 and is, therefore, closed. No mathematical system postulates that the natural numbers end at 63 or is

uninterested in what might lie beyond 63. The consequences of the closed nature of the hexagram set are reasons enough to believe that Leibniz expected more hexagram or yyGram texts to be forthcoming. The lack of texts, contemporary with the *I Ching*, that use a yyGram system for some purpose is evidence that the hexagrams in the *I Ching* are not meant to be mathematical.

Zero and Chuan

For Leibniz, the hexagram Chuan (six yin lines) represented the number 0; we see that he has written “0” above Chuan on his copy of the hexagram chart. Chinese mathematicians who lived at the time of the writing of the *I Ching* did not “invent” the zero. According to Ifrah:

We appreciate the importance of that discovery when we realize that it was made only three times in history: by Babylonian scholars, by Mayan astronomer-priests, and by Indian mathematicians and astronomers. (Although Chinese mathematicians conceived the place-value principle, the zero concept was evidently still unknown to them for many centuries: it was not introduced in China till the eighth century A. D., no doubt from India.) Ifrah, 1985:433

Leibniz may not have known this. In any case, Chuan, which produces the binary number for zero, 000000, appears without comment.

Does a yin (broken) line represent zero? Only to Leibniz. There is nothing in the *I Ching* that equates a yin line with zero or, for that matter, a yang line with one²⁰. The equivalence was made because Leibniz and Bouvet wanted it to be made. Were they correct – are zero and yin, one and yang, equivalent? There is no reason why the equivalence is not correct if, and this gets us to the crux of the problem, the two systems in which the characters appear are also equivalent. In other words, zero and yin, one and yang are equivalent if binary mathematics and the yyGram system, at least as it appears in the *I Ching*, are equivalent.

If yin and yang and zero and one are equivalent, they may be used interchangeably. Where one can be used, the other can be used. We can see

how Leibniz and Bouvet could have seized on the superficial similarity between binary numbers and the hexagrams to claim that there was equivalence. After all, the hexagrams could be reduced to binary numbers. It was plausible to say that the lines and the digits were interchangeable if one went no further in examining the yyGram system. Once the equivalence were postulated and accepted, it would be natural to assume that the systems were also equivalent. The fact that the *I Ching* is far from a mathematical text or a text that uses the hexagrams as Real Characters could easily have led Bouvet and Leibniz to believe that the “true understanding” of the hexagrams had been lost or corrupted over the centuries. They, Bouvet and Leibniz, had rediscovered the hidden meaning of the hexagrams as Fohi originally constructed them. If this is the argument that Leibniz and Bouvet used, they accepted circular reasoning in their enthusiasm.

What of Chuan? It equals 0 if interpreted as a binary number. The reader may ask whether all number systems or, at least, all sophisticated number systems need zero. Surprisingly, the answer is “no”. It is instructive to examine how the Chinese “assembled” large numbers. A large number using “traditional” Chinese numbers is a combination of multiplications and additions rather than being a place order numeral. For example, the number 7654 in traditional Chinese numbers is *qi qian liu bai wu shí sì*²¹ or $7 \times 1000 + 6 \times 100 + 5 \times 10 + 4$. Zero would be redundant in such a system. Ifrah²² 1985:365-367 shows that, by extension, traditional Chinese notation can be used to express numbers as large as 10^{32} and, theoretically, as large as 10^{4096} . Obviously, numbers as large as these go far beyond the needs of everyday living or engineering. The earliest records of Chinese numbers and mathematics occur in the oracle bones. While the signs for numbers are different, the only significant difference in the method of constructing large numbers in the oracle bones is that they use special symbols for 20, 30, and 40 that were later replaced with multiplications. The method of construction using multiplication and addition is otherwise the same in the oracle bones and in the traditional Chinese number system Ifrah 1985:347-352²³.

The traditional numerals were not the only method used by the Chinese for calculation. Rod numerals were used for calculations usually performed on a counting board²⁴. Rod numerals were used in a place-value system. The Chinese were able to perform quite sophisticated mathematical operations using rod numerals including the solution of simultaneous equations (n equations with n unknowns). Nelson illustrates the construction of “Pascal’s”²⁵ triangle using rod numerals Nelson 1993:135. Ifrah 1985:120 illustrates a portion of a counting board with three rod numeral equations one of which has the quotient zero. Here, zero is indicated by a blank square. The illustration is derived from a Han work, probably the anonymous *K’iu-ch’ang Suan-shu* (*Calculation in Nine Chapters*) Ifrah 1985:118. While this suggests that the Chinese may have had some understanding of zero by the Han dynasty, it does not tell us what that understanding was²⁶. Certainly, Han dynasty Chinese mathematicians did not feel the need to develop a special symbol for zero even though solving simultaneous equations would have more use for a symbol for zero than would mere counting.

We can feel on certain ground when we say that there was no symbol for zero as recently as the Han dynasty because, with the introduction of Indian mathematics in the eighth century A.D. along with Buddhism, Chinese mathematicians did use a symbol for zero at times with rod numerals (probably written) and traditional numerals Ifrah 1985:459-461. Our brief foray into the history and use of zero in Chinese mathematics provides us with additional proof that Leibniz did not understand the nature of the hexagrams in the *I Ching*.

Mathematical operations and the hexagrams

If the hexagrams or, more generally, yyGrams are numerals used in a mathematical system, we expect that they could be used in mathematical operations. We limit our discussion to arithmetic: addition, subtraction, multiplication, and division. Indeed, we need look no further than the *Discourse* for models of how the hexagrams might be used in basic arithmetical operations, assuming that Leibniz’s analogy between binary numbers and hexagrams holds.

First, we must confront the fact that there are no mathematical operations in the *I Ching*. This is another reminder that Leibniz and Bouvet did not base their assertion that the hexagrams were the basis (or part of) of a system of binary mathematics on actual Chinese mathematics but on appearances only. There are very limited uses of numbers in the *I Ching*. Using Shaughnessy's edition of the Mawangdui text, six and nine appear most frequently as names²⁷ rather than as numbers; six does not appear as a number at all. The following are the only numbers that appear in the received version or the Mawangdui version of the *I Ching*²⁸: 1, 3, 7, 8, 9, 10, and 300 for a total of 36 occurrences. These numbers are not evenly distributed as **Table 1** shows:

Number	Occurrences
1	4
3	20
7	2
8	1
9	1
10	5
300	1

Table 1 – Occurrences of Numbers in the *I Ching* Text

Each hexagram is associated with text that has seven parts, an introduction and a text for each line. **Table 2** shows the distribution of numbers across these texts²⁹:

Text Portion	Count
Introduction	4
Initial (bottom) line	3
Second line	7
Third line	6
Fourth line	2
Fifth line	6
Elevated (top) line	6

Table 2 – Distribution across Text Portions

The numbers are nearly evenly divided in their association with yin and yang (6 and 9) lines, as **Table 3** shows:

Line type and position	Count
Initial 6	2
Initial 9	1
6 in the second	3
9 in the second	4

6 in the third	3
9 in the third	3
6 in the fourth	1
9 in the fourth	1
6 in the fifth	5
9 in the fifth	1
Elevated 6	4
Elevated 9	2

Table 3 – Distribution across line types and positions

It is clear from **Tables 1, 2, and 3** that the *I Ching* is not a work of mathematical computation³⁰.

Giving Leibniz and Bouvet every benefit of the doubt, we should briefly inquire whether the yyGram system in general or the hexagrams in particular could be used for mathematical computation. The closed nature of the hexagram set creates many problems for mathematical operations. Using Leibniz's method for determining the binary number of a hexagram for a moment, we can see that some operations are nearly plausible. For example we can say that Chuan (binary 0) added to or subtracted from any hexagram gives us the value of the second hexagram unchanged. Perhaps Shaoguo (binary 000100 or 4) + Guimei (binary 001011 or 11) = Taizhuang (binary 001111 or 15). However, any hexagram other than Chuan that is added to Jian (binary 111111 or 63) results in a number outside the hexagram system³¹; Jian subtracted from any hexagram produces a negative number³².

Multiplication and division create other problems. It is easy to produce a number larger than can be represented by a hexagram by multiplying two hexagrams together. For example, raising a number greater than 7 to the power of 2 results in a number outside the hexagram set. Division leads us into an entirely different set of concerns. Addition, subtraction or multiplication of two integers (all hexagrams are integers) will result in another integer. Such is not the case with division³³. A simple case is dividing 2 by 3. A "fractional" hexagram would be analogous to a fraction using Arabic numerals: two numbers separated by the division sign. If the result is either outside the hexagram range or is not an integer, representation is not straightforward. There are no symbols for addition, subtraction, multiplication or division in our yyGram system or in the *I Ching*.

Leibniz briefly discusses binary arithmetical operations in the *Discourse*. His briefest discussion is about division. All he says is:

In division there is no need to tally as in ordinary calculation.

One must only see if the divisor is greater or less than the preceding remainder Cook and Rosemont 1994:137.

If an even number were divided by two, division amounts to a shift to the right. Four (0100) divided by 2 (0010) is two (0010). The 1 shifts from the fours place to the twos place. Dividing an odd number by two presents a problem that has only recently, i.e. since Leibniz's day, been solved by computer scientists³⁴. Division may result in either a decimal portion or an integer remainder, depending on how one represents the division. We have examples of neither from the *I Ching* or, indeed, from the *Discourse*. When Chinese mathematicians began to use a zero numeral with rod numbers, they were able to represent decimal fractions Ifrah 1985:393.

Hexagram sequences

Leibniz and Bouvet may not have been aware that there was more than one sequence for the *I Ching* hexagrams. The two sequences are the Fu Xi sequence, which Leibniz used, and the King Wen sequence. Not knowing that there were two sequences, Leibniz did not explain the reasons for there being different arrangements of hexagrams. There is only one arrangement of natural numbers. Any other sequence would seem "out of order". By using his method rather than the Chinese method (if, indeed, he even knew that there was a Chinese method), Leibniz saw binary numbers emerge from the hexagrams in the order that he expected them to emerge. Convinced as he was that the Chinese had forgotten the "true meaning" of the hexagrams, he did not consider any other possibility. He did not even consider the fact that the sequence of hexagrams in the square was different from the sequence of hexagrams in the circle on the chart that he used.

Table 4 shows the numbers derived from the hexagrams in the Fu Xi sequence using the Chinese reading:

0	32	16	48	8	40	24	56
4	36	20	52	12	44	28	60
2	34	18	50	10	42	26	58

6	38	22	54	14	46	30	62
1	33	17	49	9	41	25	57
5	37	21	53	13	45	29	61
3	35	19	51	11	43	27	59
7	39	23	55	15	47	31	63

Table 4 – Numbers of the Fu Xi Sequence using the Chinese Reading

The number sequence is very different than the one that Leibniz saw. Leibniz might have seen this sequence as proof that the Chinese had forgotten the mathematics of Fu Xi and reduced the hexagrams to symbols³⁵. Leibniz does not mention the King Wen sequence. Deriving numbers from hexagrams read in the Chinese fashion results in **Table 5**.

16	2	58	23	34	17	0	63
8	4	47	61	56	7	59	55
1	32	37	41	48	3	23	25
13	28	45	18	30	33	39	57
10	20	43	53	5	40	15	60
22	26	6	24	62	31	49	35
44	15	11	52	36	9	46	29
42	21	12	51	19	50	27	54

Table 5 - Numbers of the King Wen Sequence using the Chinese Reading

This sequence, too, is “out of order”. The alert reader might notice that some pairs of cells are double (or half) of the other, as we noted above. In general, the King Wen sequence does not have an obvious mathematical construction. When we look at the hexagrams, the orderly nature of this sequence becomes apparent. Take two hexagrams, A and B. B is the vertical mirror image of A. We can see a pattern for construction in the Fu Xi sequence as well, although it is a bit more complicated. If we divide each hexagram into upper and lower trigrams, we hold the lower trigram constant across each row. The upper trigrams change from column to column in a fixed order. The order is not the order of natural numbers but, rather, it is the order of the Fu Xi trigrams read in a certain way. If we read the octagon of the *ba gua* as a compass dial with south at the top, the upper trigrams of the first four columns are north, northwest, west, and southwest (the four trigrams reading from the bottom and going counterclockwise). The upper trigrams of the last four columns are northeast, east, southeast, and south (the four

trigrams reading from the bottom left and going clockwise). The lower trigrams are in the same order when read down a column³⁶.

If Leibniz noticed the order of hexagrams in the circle in his chart, he did not comment on it. It is not in a continuous order of natural numbers. It is in the order derived from the Fu Xi trigrams described above. The Fu Xi trigram order explains why the circle seems to be broken. It also demonstrates that the order of hexagrams in the Fu Xi sequence is not based on binary numbers represented by the hexagrams.

Hexagrams removed from the *I Ching*

We need to consider one final piece of evidence. If a number is isolated on a page, we still know its value. We can determine whether the character “2” is used as a value, as a name, or as a design element. The same cannot be said for hexagrams. **Figure 3** shows a Neolithic pot with three sets of six unbroken lines.



Figure 3 – Chinese Neolithic Pot

Each set of lines could be the hexagram Jian but are they? The six lines repeat three times but no other hexagram appears on the pot. In short, the hexagram is “out of context”. There is no obvious connection between the text for Jian and this pot. Are these hexagrams or merely decorations? Are these decoration supposed to be reminiscent of the hexagrams? We cannot tell just

by looking at the pot. If these were Arabic or traditional Chinese numerals or rod numerals, we would have no doubt. This suggests that Leibniz and Bouvet identified the hexagrams with binary numbers because the hexagrams were in a set. They accepted the set as a context and equated the hexagram set with the set of natural numbers or a subset of natural numbers.

Seeking and Finding

Leibniz and Bouvet found binary numbers in the hexagrams because they wanted to find binary numbers or some universal numbering system somewhere. They were actively looking for such a system because their belief in a particular structure of the world required one. Leibniz's belief was so strong that he was willing to discount 2000 years of Chinese learning. The Chinese "forgot" Fu Xi's great teachings. Leibniz did not venture what the hexagrams "meant" although he did say that, whatever they meant, it wasn't what the Chinese said they meant. For Leibniz, there was no relation between the *I Ching* texts and the hexagrams. Recall that Leibniz wanted a Universal Characteristic that was not only a language that could be read by anyone but a sort of language *in which all notions and things are nicely ordered, a language with whose help different nations can communicate their thoughts, and each, in its own language, read what the other wrote* Ariew and Garber, *ibid* (emphasis supplied). Here was a universal language that the Chinese could not properly read in their own language. When Leibniz read it in his language, it meant something entirely different from what the *I Ching* said it meant. To save the theory, either Leibniz or the Chinese had to be wrong. Since Leibniz read the text *with the aid of logic, critical thinking, mathematics and our manner of expressing thought – more exacting than theirs*. Cook and Rosemont *ibid*, the Chinese were wrong. Therefore, they had forgotten the true meaning of Fu Xi's design.

The *Discourse* is unfinished so we do not know whether Leibniz intended to tell us what the hexagrams meant, if anything, beyond mere mathematics. He may have thought that the Universal Characteristic should

be a language composed of fixed ideas with assigned unique numbers. At the very least, Leibniz oversimplified the difficulties involved in this project.

I do not argue that there are no mathematical relations between hexagrams or trigrams in the *I Ching* or in other works of Chinese philosophy. I argue that the hexagrams do not represent binary numbers as Leibniz developed them. More specifically, the hexagrams were not constructed according to the binary number system invented (he would say “reinvented”) by Leibniz. I do not argue that the Chinese could not have developed a binary number system. I argue that the Chinese did not develop a binary number system for hexagram construction. The *I Ching* and works of philosophers derived from or commenting on the *I Ching* are full of binary, i.e., two-part, relations, e.g., yin and yang, light and dark, hardness and softness. Any binary relationship used in a systematic way will produce a specific number of variations, depending on the number of two-part combinations used. In a system using six lines, in which each line can have only one of two representations, there are exactly 64 variants. One does not need recourse to mathematics, much less binary mathematics, to construct them all. All one needs is some systematic thought.

We saw that the Fu Xi hexagram sequence is directly related to the Fu Xi trigram sequence. This relation provides a complete means of construction for the hexagrams. Knowing the relation between the trigrams and the hexagrams, one can construct all 64 hexagrams without recourse to mathematics. It may be tempting to see this form of construction as an alternate to mathematical construction via binary arithmetic. Rather, it is the primary means of construction; it is the construction method that the Chinese used. The relations between numbers in the hexagram charts can be understood entirely without recourse to mathematics, although, of course, mathematics can also be used to understand the relations between the numbers in the *I Ching*.

This historical episode should caution us about “necessarily” finding Western ideas in Chinese culture (or vice versa). We need to approach the texts cautiously and humbly so that we do not misinterpret the texts the way

that Leibniz did. Some ideas may truly be universal but we cannot say for certain what those ideas may be until we test them against as many cultures as possible. Western scholars have assumed that the West has discovered the world “as it is”. Perhaps we should reconsider that: the West has discovered the world as it may be.

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¹ The copy of the Yi Jing hexagrams that Leibniz received is reproduced in Mungello, 1999:71. Note that the illustration is upside down. The same hexagram chart is reproduced as Plate II in Legge 1963.

² Rather than describing binary mathematics here, the reader should consult the references.

³ Computers keep track of things through a huge number of switches. Originally, these switches were large and visible; today, they are microscopic and very fast. All switches that form computer memory have two states, equivalent to on and off, which can be represented by a binary numbering system as 1 and 0 respectively.

⁴ The great twentieth-century logician Kurt Gödel also proposed a system for calculating knowledge. The system often referred to as “Gödel numbering” applies to any expression that can be placed into logical notation, i.e., the notation of predicate calculus. This system is substantially different in design from Leibniz’s system. The interested reader is referred to Gödel, 1992 and Nagel and Newman, 1958.

⁵ In computer jargon, a “killer app” is a program (an application) that everyone has to have and which replaces other programs that treat its subject matter (subject matters like word processing or web browsing). Web browsers were “killer apps” when they were first introduced.

⁶ Fohi is Fu Xi.

⁷ Reason also had a theological component. One might even say that reasoning was a branch of theology for Leibniz. In his *Theodicy*, he says:

I assume that two truths cannot contradict each other; that the object of faith is the truth God has revealed in an extraordinary way; and that reason is the linking together of truths, but especially (when compared with faith) of those whereto the human mind can attain naturally without being aided by the light of faith... Reason, since it consists in the linking together of truths, is entitled to connect also those wherewith experience has furnished it, in order thence to draw mixed conclusions; but reason pure and simple, as distinct from experience, only has to do with truths independent of the senses. Leibniz, 1985:73.

⁸ Remember, the illustration in Mungello is upside down.

⁹ It is not clear from his writings whether he knew that the Chinese read the hexagrams from the bottom up. The fact that reading the hexagrams from the top down created an ordered sequence of digits could have reinforced Leibniz’s belief that the contemporary Chinese did not properly understand the hexagrams or Fu Xi’s purpose for developing them.

¹⁰ I use the hexagram names from Shaughnessy, 1996.

¹¹ We need six digits to represent 6 lines. Note that in the binary notation developed by Leibniz, the leading 0 would be dropped. We will continue to use six digits because hexagrams, by definition, have six lines.

¹² The fact that number doubling occurs with Wei Ji is due to the alternation of yin and yang lines in the hexagram.

¹³ I will argue later that there is a correspondence that is not obvious through Arabic numerals.

¹⁴ It is not important whether we use 0 and 1 or a broken and unbroken line to fill places since, for Leibniz, they are equivalent. Both notations are subject to the same rules for Leibniz. The yin and yang lines are numerals just as *yi*, *er*, *san* etc. are numerals.

¹⁵ Recall that the rotations that we have discussed are 90° rotations. We may say that a Leibniz rotation (90° clockwise) followed by a Chinese reading (effectively a 90° counterclockwise rotation) of the resulting binary number results in a 180° clockwise rotation of the hexagram.

¹⁶ By use of the abbreviation *etc.* at the end of two columns showing the correspondence between binary and Arabic numbers, Cook and Rosemont, 1994:73.

¹⁷ In the *Discourse*, Leibniz mentions other advantages provided by binary numbers, Cook and Rosemont, 1994:136-138. These advantages amount to simplified arithmetical operations. Indeed, this is one of the reasons that the binary number system is used in modern computers.

¹⁸ This is precisely what we see with trigrams and hexagrams so there is no basis for assuming that it would be different with *yyGrams* of any other size.

¹⁹ We will not discuss whether trigrams should be seen as the basic unit of a yyGram system. A reasonable argument, supported by the traditional construction of hexagrams, could be made that all expansions of a yyGram system should be by trigrams rather than by single lines. It is also possible to argue that a two line group should be considered the basic element.

²⁰ The Yi Jing refers to a broken line as a 6 and a solid line as a 9.

²¹ The numerals *qian*, *bai*, and *shi* may be omitted without any loss of clarity.

²² Unfortunately, Ifrah does not provide us with dates. It is highly unlikely that the Zhou-era Chinese needed numbers as large as 10^{4096} .

²³ Number systems that use the same multiplication and addition principle as the traditional Chinese system developed in Pharonic Egypt. The Egyptian system did not have a symbol for zero. Joseph notes that *No difficulties arise from not having a symbol for zero in this numeral system*. Joseph 1993:98. Interestingly, Leibniz briefly referred to this principle in the *Discourse* in the work of a contemporary mathematician named Erhard Weigelius (Weigel) but does not refer to the Chinese system. Cook and Rosemont 1994:135.

²⁴ A counting board most commonly looks like a chessboard, i.e., a board with rows and columns. Counters are placed in the squares made by the intersections. Counting boards were used by many cultures throughout the world.

²⁵ I quote *Pascal* despite the fact that the most common name of this object is *Pascal's triangle* because, while Pascal developed the triangle in 1653, the Chinese version reproduced at Nelson 1993:135 is from 1303. Ifrah 1985:396 cites the source of this figure as the *Ssu Yuan Yü Chien* by Chu Shih-Chieh.

²⁶ Since rod numerals were three-dimensional objects, the empty square could simply have been the absence of rods. The mathematician need not necessarily have thought of this as a number. This bears further study. The illustration from Ifrah mentioned above also shows that rod numerals could represent positive and negative numbers by different colors for each. Why there is no separate symbol for zero is an open question but it is clear that there was none.

²⁷ Six and nine represent yin and yang respectively.

²⁸ I have not considered the commentaries at all.

²⁹ **Table 2** does not include a count of the numbers that describe the placement and “type” of the line text, e.g., “six in the second”.

³⁰ Curiously, one hexagram is named for mathematical computation, *Suan*.

³¹ Another way of saying this is that Jian added to any hexagram except Chuan places us outside the hexagram system.

³² As we have seen from the rod numeral example, this can be handled by coloring the hexagram. Let us say that a red hexagram is negative. This strictly places us outside the hexagram system of the Yi Jing but still permits us to represent certain numbers (-1 to -63) with hexagrams. Perhaps a hexagram could be an absolute value.

³³ For a complete discussion, see Vorob'ev.

³⁴ Rather than burden the reader with discussions of mantissa, binary-coded decimal notation, and the like, the interested reader is referred to the literature on this subject.

³⁵ There is an interesting mathematical relation in this sequence. Add 8 to the number in column 1 to get the number in column 5 of the same row. Each row contains all subsequent numbers increased by 8 if counted in this sequence: 5,3,7,2,6,4,8. We will not discuss this further but it may have something to do with the Fu Xi trigrams.

³⁶ The King Wen sequence seems to have an order based on the King Wen trigrams. It will not be examined further in this paper.